

Biology and Larval Feeding Habits of Coexisting Hydroptilidae (Trichoptera) from a Small Woodland Stream in Northeastern Ohio

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ABSTRACT *Hydroptila consimilis* Mosely, *Ochrotrichia spinosa* (Ross), and *O. wojcickyi* Blickle exhibited a close association with the appearance of the filamentous chlorophyte *Cladophora* (April to June) in a small woodland stream in northeastern Ohio. Laboratory rearings showed that these microcaddisflies displayed similar life cycles and exhibited hypermetamorphosis. *H. consimilis* mandibles are specialized for piercing individual cells within filaments of *Cladophora* to facilitate removal of cellular contents, whereas *Ochrotrichia* spp. have robustly cusped mandibles suited for piercing *Cladophora* and scraping diatoms from benthic substrates. Proportional similarity analysis of videotaped behavior illustrated that the feeding habits of third-instar *Ochrotrichia* exhibited a low similarity (0.44) to the other instars tested (*Ochrotrichia* fifth and *Hydroptila* first, third, and fifth) because of their frequent consumption of diatoms (17% of feeding efforts); all other instars tested were highly similar to each other (0.72-0.86). This division of trophic resources (i.e., differential use of diatoms versus *Cladophora*) minimized niche overlap between the two genera. All instars of *H. consimilis* and *Ochrotrichia* spp. fed heavily on the apical (therefore the smallest) cells of *Cladophora* filaments, and the time required to consume cells decreased significantly as larvae matured ($P < 0.05$). Larval Hydroptilidae are well adapted morphologically and behaviorally to consume *Cladophora*, and these algal filaments appear to represent a heterogeneous food source just as terrestrial host plants are for the array of phytophagous insects that use them.

KEY WORDS Hydroptilidae, niche overlap, *Cladophora*, grazing, streams

MANY SPECIES OF invertebrates that share seemingly similar resource requirements live in sympatry. Competitive interactions have been described for aquatic insects that overlap in their spatial and trophic niches (e.g., McAuliffe 1984, Hart 1985a). However, close scrutiny of invertebrate life histories has shown that niche partitioning frequently occurs, thus explaining the coexistence (Price 1997). Among aquatic and semiaquatic invertebrates, ephydrid flies (Diptera) (Foote 1977, 1979, Zack and Foote 1978, Connell and Scheiring 1981), caddisflies (Trichoptera) (Mackay and Kalff 1973, Malas and Wallace 1977, Hildrew and Edington 1979, Teague et al. 1985, Harding 1997), gastropods (Schmitt 1996), and others have been shown to divide trophic and microspatial niches to facilitate coexistence. Furthermore, populations of benthic macroinvertebrates living in streams that experience unpredictable disturbances (e.g., spates) may not reach the numbers required to exclude competing heterospecifics sharing similar resource requirements (Connell 1978, Reice 1994). The classic work by Paine (1969) showed a reduction in species richness of the intertidal invertebrate community consumed by starfish when competitively superior species excluded heterospecifics when their predator was removed. Clearly, no one ecological process can account

for all situations of intimate sympatry observed in nature.

Trichoptera is a diverse order of holometabolous insects, and most larvae are aquatic throughout their existence. Microcaddisflies (Hydroptilidae) represent the second largest trichopteran family in North America, Greenland, and Mexico with >300 species (Morse 1993); most species are thought to exist in streams, but the number of known larvae is only a small fraction of this family's species richness (Wiggins 1990, 1996). Although studies of *Leucotrichia pictipes* (Banks) (Shapas and Hilsenhoff 1976, McAuliffe 1982, 1984, Hart 1985a, b) and *Dibusa angata* Ross (Resh and Houpp 1986) from the eastern United States gave detailed information on the biology and life histories of these fascinating species, these genera are not representative of the other hydroptilid taxa found throughout North America leaving a dearth in our knowledge of the larval biology of this large family. The biology of Palaearctic (Nielsen 1948, Lepneva 1964), Oriental (Ito and Kawamura 1980, Ito 1998), and Australian (Wells 1985) hydroptilid faunas are somewhat better understood.

Despite being common stream insects exhibiting interesting biology and life history characteristics, no comparative studies have been conducted to explain their coexistence. We initiated a study of the Hydroptilidae from a small woodland stream in northeastern Ohio (USA) to shed light on their general biology and

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life history through field studies and laboratory rearings. We attempt to explain how their coexistence is facilitated by morphological, behavioral, and life history characters. We also investigated their close ecological association with the filamentous alga *Cladophora* (Chlorophyta) as a host alga.

Materials and Methods

Study Site. Bixon Creek is a third order stream in West Branch State Park, Portage County, OH. It flows through a beech/maple (*Fagus grandifolia* Ehrh./*Acer rubrum* L.) forest, and is well shaded during the late spring and summer months. Five riffles, situated near 41° 07' 11" N, 81° 09' 31" W, were exposed to direct sunlight through a break in the forest canopy. *Cladophora* grew in lush mats on riffle rocks under this break, whereas other nearby riffles lacked prolific growths of algal filaments. Larvae of the microcaddisflies *Hydroptila consimilis* Mosely, *Ochrotrichia spinosa* (Ross), and *O. wojcickiyi* Bickley co-occur among the algal masses.

The five riffles used during this study differed in size, although each had a mixture of gravel and cobble >90% of which was <15 cm in diameter. Each riffle was designated 1–5, beginning with the one farthest upstream; ≈55 m of channel length separated the first and last riffles. The lengths of the riffles were ≈10, 8, 4, 6, and 6 m, respectively. The width of the channel was ≈3.5 m maximum, but both the length and width varied throughout the study period because of fluctuating water levels. The channel narrowed to <1 m at riffle 3, which had several large boulders within the channel. The banks of the stream were up to 1.25 m high.

Phenology of Bixon Creek Hydroptilidae. To obtain physical data on the larval habitat, water temperature and a visual estimate of the percent cover of *Cladophora* were recorded at each riffle in 1997. Flow rates were determined using an American Oceanics analog flow meter; two measurements were taken in each riffle, ≈0.5 m from the riffle head and foot in the center of the channel. Riffles 1, 2, 4, and 5 were sampled once per collecting date for immature stages using a 9.5 by 9.5-cm quadrat placed on randomly selected masses of *Cladophora*. Algal filaments, sediment, and periphyton within the quadrat were scraped or collected with forceps from the riffle rock(s), and placed into a 0.45-liter jar with stream water. Riffle 3 was not sampled for immature stages because sampling would have been catastrophic for the benthic populations in this small area, and was used for percent coverage, flow, and temperature measurements only. Because of the close association of larvae with *Cladophora*, quantitative samples were obtained only when algal mats were present.

Quantitative samples were returned to the laboratory, the contents of each jar were placed in separate shallow pans, and larval and pupal hydroptilids were collected and preserved in 70% ethanol. For each sampling date, numbers of immature stages from each riffle were recorded, the larvae from all riffles were

pooled, and a random subsample of 50 individuals was obtained for head capsule measurement and instar determination. On 8 May, <50 larvae were collected, therefore all larval head capsules were measured.

A single fluorescent light trap was run at riffle 3 every 5–8 d, from 31 May to 25 July 1997 to obtain data on flight periods for Bixon Creek Hydroptilidae.

Laboratory Rearings. During the late spring and summer months of 1995–1997, larvae were collected from riffle rocks with forceps, placed in jars of stream water, and returned to the laboratory in a cooler with ice. Small riffle rocks and algal masses were also obtained. Larvae, rocks, and algae were placed in aerated rearing chambers (Keiper and Foote 1996) with stream water to acquire emergent adults for species determination. Chambers were kept at 19–21°C and a photoperiod of 16:8 (L:D) h maintained with fluorescent lights. Laboratory rearings were initiated to obtain life cycle data using field-collected adult females taken in June 1996 and 1997 following the methods of Keiper et al. (1998a) and Keiper and Foote (1999).

To obtain information on feeding habits, larvae were observed usually daily at 6–50× with a dissecting microscope. Larvae were videotaped to facilitate repeated observations. A stage was constructed allowing observers to manipulate rearing dishes in three dimensions thereby not disturbing the larvae. A fiber-optics light source set at the lowest intensity needed for observation was used to illuminate the specimens. Videotapes were reviewed in slow motion and normal time with a video cassette player.

Gut contents of *Ochrotrichia* and *Hydroptila* larvae taken from Bixon Creek riffle rocks and from mats of *Cladophora* were examined by dissecting the digestive track from the abdomen and thorax. The guts remained intact when removed from the body and were macerated in a drop of glycerol on a glass microscope slide to which a cover slip was added. Diatoms were counted and other materials were described using a compound light microscope. If the diatoms on a slide were too numerous to count, the number of diatoms in one field was counted and the number of fields with same density (approximated visually) as the one counted was recorded to provide an estimate. Numbers of diatoms were log ($n+1$) transformed for each sample to facilitate statistical analysis with a Student *t*-test.

Behavioral Analysis. Quantitative video recordings were used to obtain data on interspecific differences and ontogenetic shifts in the behavior of larvae encountered among filaments of *Cladophora* in Bixon Creek. Larvae of *H. consimilis* were obtained by rearing larvae from eggs obtained from field-collected females. Because of the difficulty of acquiring adequate numbers of *Ochrotrichia* by rearing eggs, larvae of this genus were collected from Bixon Creek. Larvae of *O. spinosa* and *O. wojcickiyi* could not be separated by morphological characters. However, rearing field-collected larvae during 1995–1997 showed that *O. spinosa* consistently represented a small proportion of the Hydroptilidae from Bixon Creek (J.B.K., unpublished data).

Table 1. Definitions of behaviors recorded

Behavior	Definition
Crawling/searching	Larva actively moves among algal masses, and is not executing feeding behaviors.
Feeding apical cell	Larva biting into or removing the contents of a <i>Cladophora</i> cell situated at the distal end of a filament.
Feeding subapical cell	Larva biting into or removing the contents of a <i>Cladophora</i> cell situated next to an apical cell.
Feeding basal cell	Larva biting into or removing the contents of a <i>Cladophora</i> cell which is positioned neither apically or subapically.
Feeding diatoms	Larva places mandibles on outer surface of <i>Cladophora</i> cell and articulates them rapidly to consume epiphytic diatoms.
Other	All behaviors not related to feeding habits, including periods when larvae are not observable, pausing (resting), grooming, and withdrawing into case (fifth instars only).

Larvae were placed individually in petri dishes of filtered stream water with a thin lawn of *Cladophora* covering $\approx 50\%$ of the dish bottom. Algal filaments were field-collected and gently rinsed with distilled water. After acclimating to the new dish for at least 0.5 h, first-, third-, and fifth-instar *H. consimilis* and third- and fifth-instar *Ochrotrichia* sp. were recorded. Ten individuals of each instar were recorded for 20 min each, for a total of 1,000 min of videotape.

During video playback, behaviors were recorded for each individual on sheets with 1,200 1-s intervals, for a total of 12,000 1-s observations per instar examined. To determine if significant differences exist in the feeding ecology of the two genera, behaviors recorded were grouped into the six categories of crawling/searching, feeding on an apical cell, feeding on a subapical cell, feeding on a basal cell, feeding on epiphytic diatoms, and others (Table 1). The mean number of bouts and proportion of time spent in each behavioral state were calculated for each larva, and then pooled among individuals of the same instar and species. Parametric (Student *t*-test or one-way analysis of variance (ANOVA) coupled with Tukey posthoc test) or nonparametric tests (Mann-Whitney rank sum Test or Kruskal-Wallis one-way ANOVA on ranks coupled with Student-Newman-Keuls posthoc test) (Zar 1996) were used where appropriate to determine if the proportion of time or number of bouts differed significantly among the instars tested.

To determine if larvae exhibited ontogenetic shifts in their feeding efficiency (defined as the time to consume the contents of one cell of *Cladophora*) and to obtain data illustrating that *Cladophora* is a heterogeneous food source for hydroptilid larvae, the mean time to consume each cell type for each instar/genus tested was calculated. Simple linear regression with replication (Zar 1996) was executed by plotting head capsule width (X) against $\log_{10}$ transformed mean time to consume each cell type (Y) to test the hypothesis that time to consume the contents of a cell is dependent on head capsule width (i.e., instar).

To garner data on resource partitioning, data were pooled for the third and fifth instars for each genus and compared with a *t*-test; the data on *H. consimilis* first instars were not used because data for first-instar *Ochrotrichia* sp. were not obtained. First, the mean number of *Cladophora* cells consumed by each genus was compared, and the null hypothesis was the total number of cells consumed by *H. consimilis* is less than or equal to the number of cells consumed by *Ochrotrichia*. Second, the proportion time spent grazing epiphytic diatoms from filaments of *Cladophora* was compared, the null hypothesis being that the mean proportion of time spent grazing diatoms by *Ochrotrichia* is less than or equal to the proportion of time spent grazing diatoms by *H. consimilis*.

The number of feeding bouts executed by third ($n = 10$) and 5th ($n = 10$) instars of each genus was combined, and the total number of bouts executed by individuals of each genus ($n = 20$ for each genus) was pooled. Proportional similarity was calculated using the following formula outlined by Price (1997) to measure the degree of overlap in trophic resource utilization:

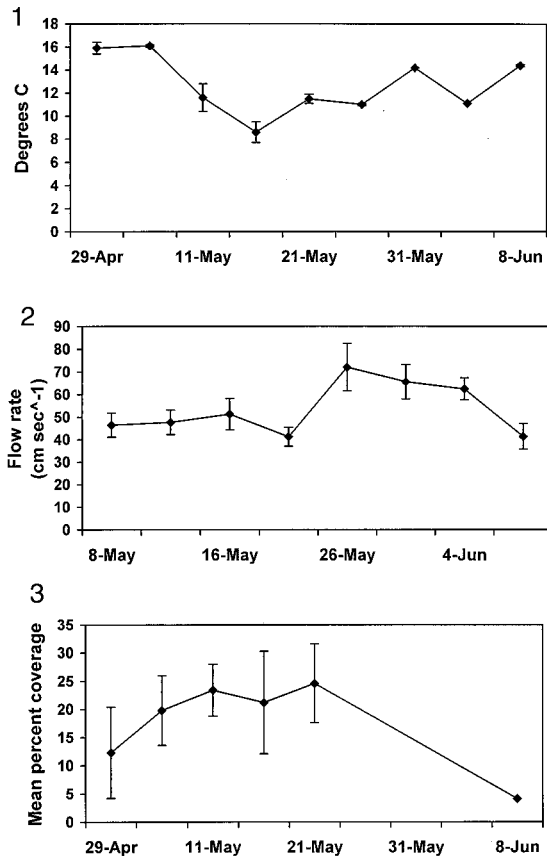
$$PS = 1 - 0.5 \sum |p_{ij} - p_{hj}|,$$

where P = the proportion of each resource (j) used, and i and h = the two genera tested. A value of 0.0 indicates no similarity, whereas 1.0 indicates complete similarity.

Results

Phenology of Bixon Creek Hydroptilidae. Water temperatures are given in Fig. 1. Flow rates were ≈ 40 – 50 cm s^{-1} , but increased to ≈ 60 – 70 cm s^{-1} between 26 May and 4 June because of precipitation (Fig. 2). The mean percent coverage of riffle rocks by *Cladophora* peaked at ≈ 20 – 30% on 21 May (Fig. 3). No data were available for 26 May, 31 May, and 4 June because of the high and turbid waters. On 8 June, $<5\%$ of the riffle area was covered by *Cladophora*. After 8 June, only scattered patches of short filaments were observed ($<1\%$ coverage).

Although it is impossible to separate the first two instars of *Hydroptila* and *Ochrotrichia* using morphological characters, instar determination for all field-collected larvae was possible using measurements obtained during laboratory rearings. The first hydroptilid larvae were encountered on 8 May; approximately half were second instars, and the remaining consisted equally of first and third instars (Fig. 4). No larvae were encountered before 8 May. The numbers of larvae collected peaked on 31 May with an approximate mean number of 190 larvae per 9.5 by 9.5-cm quadrat ($>21,000$ larvae m^{-2} of *Cladophora*-covered substrate). There was a steady decline as June progressed (Fig. 5). The proportion of larvae in each stadium shifted toward $>90\%$ of the larvae being fifth instars in late May; by 31 May, all larvae collected were in the fourth or fifth stadium (Fig. 4). Pupae first appeared on 26 May, peaked 31 May, and declined steadily thereafter (Fig. 5).



Figs. 1-3. Bixon Creek (mean $\pm$ SE) water temperature, current speed, and percent coverage by *Cladophora* filaments, 1997.

Adult flight periods for 1997 were 23 June to 7 July for *H. consimilis* and *O. wojcickiyi*, and 23 June to 14 July for *O. spinosa*. Qualitative light trapping from 1995 to 1996 illustrated that adults of each species emerged as early as late May, but unseasonably cool nights during 1997 hampered adult collection. Alternatively, there was a nearly 50% decrease in water temperature from 8 to 16 May (Fig. 1), which may have slowed larval development.

Morphology and Biology of *H. consimilis*. Head capsule width for each instar is given in Table 2. The mean factor of increase was 1.23 (range, 1.13–1.32); smallest factor of increase between first and second instars, with the largest between the third and fourth. The mandibles are asymmetrical; the left has a scalloped inner edge with the right being pointed; those of the fifth instar with three setae on posterolateral corner of left and one on the posterolateral corner of right (Fig. 6A). The sclerites of first two instars are colorless, with the mandibles exhibiting a slight yellow color; sclerites of the third instar darkened slightly, and those of the fourth instar are brown. Fifth-instar sclerites are dark brown with hyaline ring around the eye. Larvae are otherwise identical to those illustrated by Ross (1944) and Wiggins (1996).

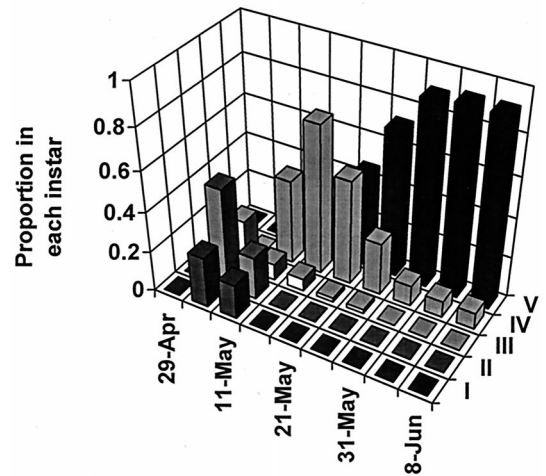


Fig. 4. Proportion of larvae in each instar at Bixon Creek ($n = 50$ for each collection date), based on head capsule widths obtained during laboratory rearings.

Four females laid 29–140 eggs ($\bar{x} = 100.3 \pm 51.5$) which measured on average 0.09 by 0.07 mm ($n = 15$). Most eggs adhered to each other in either roughly spherical masses or chains, but some separated from the masses and floated individually on the water surface. The chorion was colorless and transparent, and the embryo was dull gray and grainy in texture. The incubation period was 7–14 d.

Instars one through four consumed individual cells of *Cladophora*. The small larvae moved along algal filaments until they encountered cells small enough to pierce. To obtain the protoplast from a cell, individual larvae grasped cells with their foretarsi and positioned their heads approximately perpendicular to the filaments. Larvae clutched the cells with their mandibles and attempted to pierce the cell walls. During a bite, the left mandible maintained a stable grip while the right mandible adducted acutely; this action was facilitated with a counterclockwise movement of the head (when viewed dorsally) $\approx 30^\circ$. During a bite, the tip of the right mandible gouged the cell wall. Young larvae often delivered 10–20 bites to a cell until the right mandible pierced the cell wall, permitting access to the protoplast.

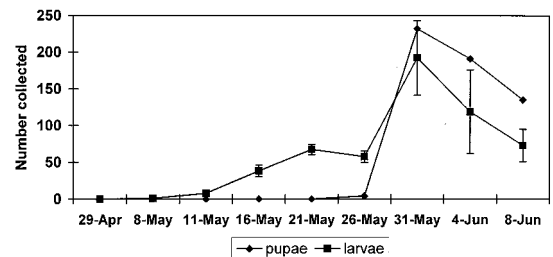


Fig. 5. Mean number of larvae (± 1 SE) and total number of pupae collected at Bixon Creek, 1997.

Table 2. Head width range, median, and factor of increase for larvae of *H. consimilis* and *O. wojcickyi*

<i>H. consimilis</i>				<i>O. wojcickyi</i>			
Instar	Head width range, mm	Head width median, mm	Factor of increase ^a	Instar	Head width range, mm	Head width median, mm	Factor of Increase ^a
I	0.98–0.105	0.102		I	0.068–0.110	0.089	
II	0.113–0.120	0.116	1.13	II	0.120	0.120	1.35
III	0.135–0.150	0.143	1.23	III	0.150–0.170	0.160	1.33
IV	0.180–0.195	0.188	1.32	IV	0.180–0.195	0.188	1.18
V	0.225–0.240	0.233	1.24	V	0.210–0.240	0.225	1.20
Mean factor of increase			1.23				1.27

^aObtained by dividing head width median of the current instar by that of the previous instar.

After puncturing the cell wall, larvae quickly pressed their mouths to the break and received the protoplast, which spewed outward from turgor pressure. First instars ingested some of the cellular material, then moved their heads away allowing a flocculent green mass to form at the break. Larvae used their forelegs to brush excess protoplast from their mouthparts. Larvae were never observed feeding on the flocculent masses of protoplast. Larvae that later encountered masses similarly ignored them, despite seemingly tasting the masses before moving on. Later instars (2–4) fed in a manner similar to first instars, but these larger larvae physically removed protoplast from cells with an apparent sucking action to drain cells. Flocculent masses of protoplast were rarely observed to form as a result of feeding by these later instars.

Final instars pierced cells relatively easily, requiring only several bites to gain access to the cellular contents. Large cells with diameters approximately equal to a larva's head capsule width required ≈10–12 bites to pierce. Normally, fifth instars completely drained cells, although larvae pierced the largest cells more than once to obtain the entire volume of protoplast. Gut contents analysis (*n* = 12) revealed that diatoms were rarely present in guts from field-collected larvae, but instead amorphous green material (presumably *Cladophora* protoplast) was present.

Only fifth instars constructed cases, beginning within 24 h of molting; a small ring of detrital material bound together with oral secretions was formed around the abdomen, and sand grains and miscellaneous debris were added to lengthen the case. Both ends were open, and larvae frequently reversed position within their cases facilitating a rapid change in direction. Completion of the purse-like case occurred

in 24–48 h, resembling that illustrated by Wiggins (1996).

Larvae attached their cases to a variety of solid substrates, including the rocks within rearing chambers and the wall and floor of the rearing chamber. Field observations at Bixon Creek showed that larvae attached their cases to ruffle rocks, often near, but never on, algal filaments. Pupal cases appeared to be aggregated.

The duration of the five instars under laboratory conditions was 3–4, 3–4, 2–3, 3–4, and 7–9 d, respectively, and the pupal period was 9–10 d. The total duration of the immature stages (egg, larva, and pupa) was 34–48 d, with a larval/pupal period of 27–34 d.

Morphology and Biology of *O. wojcickyi*. Head capsule width for each instar is given in Table 2. The mean factor of increase was 1.27 (range, 1.18–1.35); smallest factor of increase between third and fourth instars, largest between first and second. The mandibles are asymmetrical, the left bearing three robust cusps and bristles on inner margin, whereas the right is robustly cusped along inner margin and lacking bristles; those of fifth instar with two setae on posterolateral corner of left, and the right lacking setation (Fig. 6B). The sclerites of the first two instars are colorless, but the mandibles exhibited a slight yellow color; sclerites of third instar were darkened slightly, and those of the fourth brown. Fifth-instar sclerites are dark brown with a hyaline ring around the eye. The anterolateral corner of metathorax is strongly produced in mature larvae only (i.e., those >3 d old and with abdomens distended); otherwise identical to those illustrated by Wiggins (1996).

A single female deposited eggs in the laboratory. Individual eggs were ovoid, measured 0.180 by 0.105

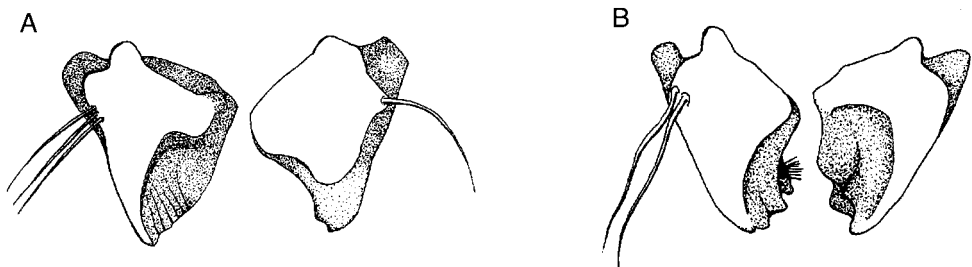


Fig. 6. (A) Mandibles of *H. consimilis*, ventral view. (B) Mandibles of *O. wojcickyi*, ventral view.

mm, and were transparent except for a central white area having a grainy texture. A thin gelatinous material covered the eggs and held them together. Developing embryos were observed, and eye spots were evident 4–6 d after being laid. Occasionally, newly hatched larvae did not escape the chorion completely, or became trapped in the gelatinous matrix where they died.

Larvae consumed the contents of individual cells of *Cladophora* and also scraped diatoms from rock substrates as well as epiphytic diatoms growing on filaments of *Cladophora* during laboratory observations. Some early instars collected from Bixon Creek consumed elliptical adnate diatoms by grabbing the ends of the frustule with the apices of their mandibles, prying one end off the cell wall and consuming it. Older larvae moved along *Cladophora* filaments and moved their heads laterally while articulating their mandibles. They ingested erect diatoms and detrital material, often pausing at areas having dense accumulations of diatoms.

Fifth instars did not consume epiphytic diatoms often during laboratory observations and appeared to focus their efforts on piercing *Cladophora* cells, although some were observed occasionally to strip adnate forms from the algal filaments. Field observations revealed that mature larvae on rock substrates faced upstream while moving their heads laterally. They appeared to break up the biofilms with their robust mandibles in this manner (see below). Gut contents analysis of larvae revealed from 6 to 4,000 diatoms, with a mean of 658.2 per larva taken from rock substrate ($n = 12$) and a mean of 803.9 per larva taken from *Cladophora*. The log ($n+1$) transformed data from rock substrate (mean = 2.42 ± 0.98) and *Cladophora* (mean = 2.34 ± 0.92) were nearly identical (not significant, $P = 0.81$).

All instars consumed the contents of individual cells of *Cladophora* in the laboratory. Early instars appeared to feed on the smallest cells within filaments. Larvae broke through the cell walls by delivering multiple bites to the cell's surface. After breaking through, larvae appeared to chew on the exposed protoplast while ingesting it.

Fifth instars fed on a wide variety of cell sizes in a manner similar to early instars. Larvae rocked their heads vertically while chewing at cell walls, and usually obtained the contents with relatively few bites. Small cells were drained easily of their contents, but *Ochrotrichia* lacked the ability to drain the contents of larger cells as was seen in *Hydroptila*; larvae frequently broke the cell wall in two or three places to consume the entire contents.

Larvae constructed cases similar in shape to those of *Hydroptila*. One difference was the frequent addition of algal filaments to the case by *Ochrotrichia* larvae, which bit through filaments of *Cladophora* to acquire short strands and attached them to the ends of cases. Some early instars collected in nature and reared on monocultures of *Cladophora* occasionally used algal filaments and oral secretions only as case material after reaching the fifth stadium. Larvae frequently reversed

position in their cases in a manner identical to *H. consimilis*.

The case of one specimen of *O. wojcickyi* collected from Bixon Creek in 1997 had a dorsal anchor stone (Ross 1944). Larvae were often seen on the upper surfaces of rocks where an anchor stone might increase the weight, reducing the chance of the larva being washed downstream. However, when observed within algal mats in the laboratory, the larva appeared hampered by the stone when it tried to penetrate thick masses of filaments.

The duration of the five stadia under laboratory conditions was 4–5, 4–6, 2–4, 3–6, and 12–15 d, respectively, and the pupal period was 9–15 d. The total duration of the immature stages (egg, larva, and pupa) was 45–63 d, with a larval/pupal duration of 34–51 d. Pupation habits of *O. wojcickyi* appeared to be identical to those of *H. consimilis*.

Because field-collected larvae and adult females of *O. spinosa* and *O. wojcickyi* cannot be separated, and because of difficulties inducing *Ochrotrichia* females to oviposit, biological details of *O. spinosa* could not be obtained. The morphology and phenology of *O. spinosa* appears to be essentially identical to *O. wojcickyi*, therefore we infer that the larval feeding habits and life history are similar.

Behavioral Analysis. First instars of *H. consimilis* spent significantly more time searching/crawling than any other instar tested for both genera (Kruskal-Wallis one-way ANOVA on ranks, $P = 0.05$). All other instars/genera spent statistically equal time exhibiting this behavior (Fig. 7). All instars/genera consumed relatively fewer subapical cells than either apical or basal, but only first and fifth instar *H. consimilis* (Kruskal-Wallis one-way ANOVA on ranks, $P = 0.05$, one-way ANOVA, $P = 0.01$, respectively) and third-instar *Ochrotrichia* (Kruskal-Wallis one-way ANOVA on Ranks, $P = 0.03$) consumed significantly fewer subapical cells; third-instar *Ochrotrichia* consumed significantly more basal cells than the other two cell types (Fig. 8). The pooled data for *H. consimilis* and *Ochrotrichia* third and fifth instars indicated that *Hydroptila* larvae consume more *Cladophora* cells per unit time than *Ochrotrichia* larvae (t -test, $P = 0.01$). Conversely, *H. consimilis* never scraped diatoms, whereas *Ochrotrichia* third and fifth instars consumed diatoms for ≈ 17 and 1% of the total time observed, respectively (Fig. 9).

Simple linear regression illustrated that significant negative relationships exist between head capsule width (instar) and the time to consume a cell type for *H. consimilis* attacking all cell types (apical, $R^2 = 0.82$, $P < 0.001$; subapical, $R^2 = 0.45$, $P = 0.02$; basal, $R^2 = 0.84$, $P < 0.001$), and for *Ochrotrichia* consuming apical ($R^2 = 0.81$, $P < 0.001$) and basal cells ($R^2 = 0.81$, $P < 0.001$). No significant trend was detected for *Ochrotrichia* consuming subapical cells ($R^2 = 0.11$, $P = 0.35$).

Results of proportional similarity calculations (Table 3) illustrate that both genera are highly similar (0.86) in the proportion of each trophic resource consumed when data are pooled for third and fifth instars. However, when individual instars of each genus are

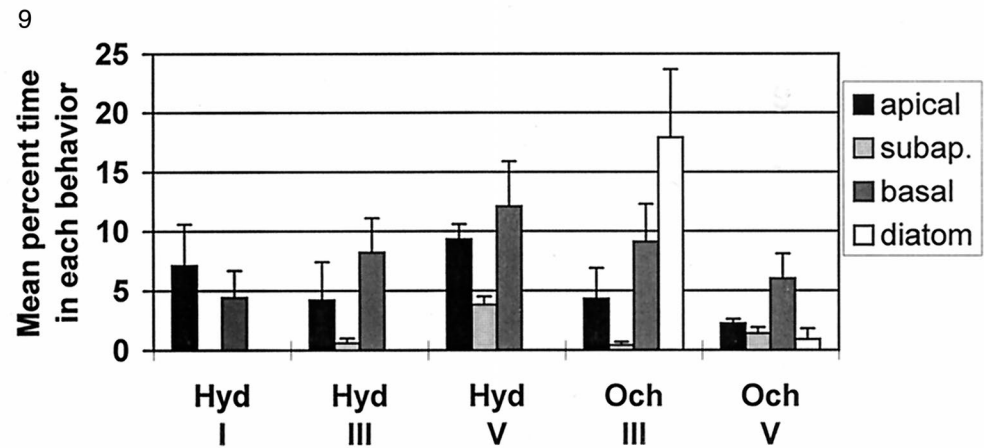
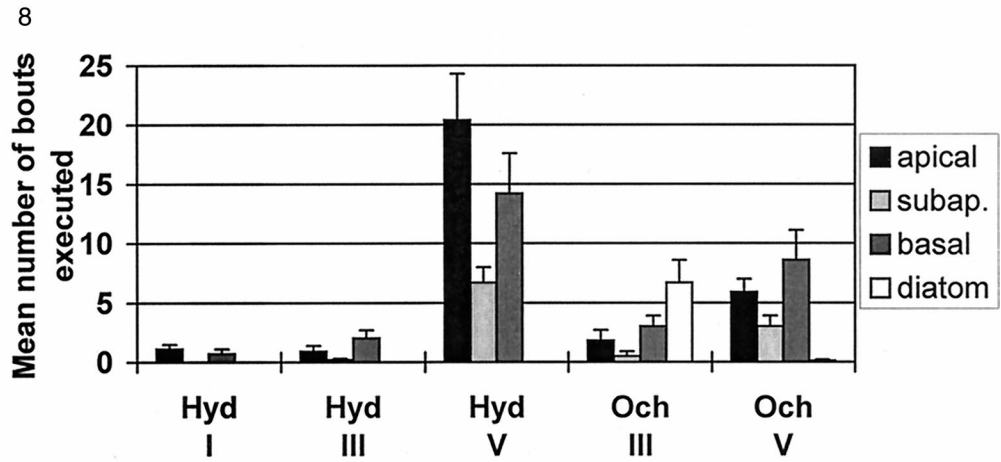
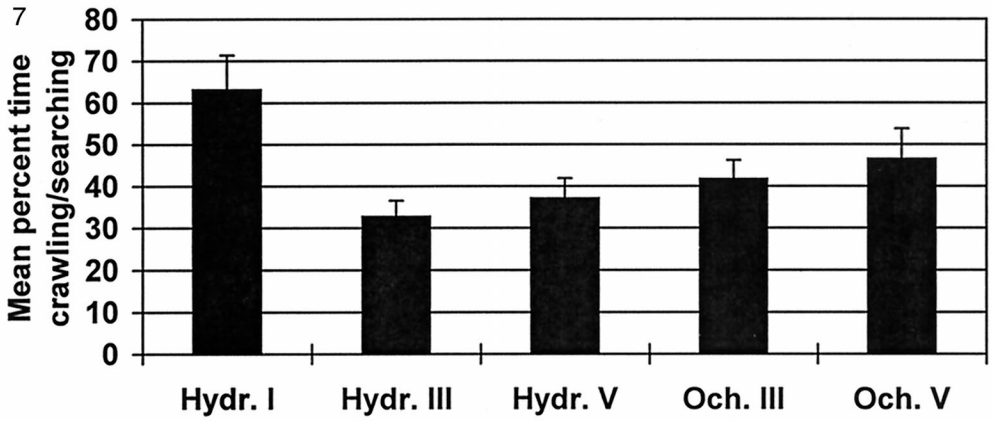


Fig. 7. Mean percent time (± 1 SE) spent in the crawling/searching behavior for each instar tested. Hydr., *H. consimilis*; Och., *Ochrotrichia* sp.

Fig. 8. Mean number of bouts executed (± 1 SE) consuming apical, subapical, and basal *Cladophora* cells and diatoms for each instar tested.

Fig. 9. Mean percent time (± 1 SE) spent consuming apical, subapical, and basal *Cladophora* cells and diatoms for each instar tested.

Table 3. Total number of feeding bouts on each food resource (n = 716) executed by 3rd and 5th instars from 20-min feeding trials

Food type	H _{III}	O _{III}	H _V	O _V	H _{pooled}	O _{pooled}
Apical cells	9	18	181	57	190	75
Subapical cells	2	5	67	30	69	35
Basal cells	20	30	142	87	162	117
Diatoms	0	67	0	1	0	68
n	31	120	390	175	421	295
PS	0.44	0.44	0.86		0.86	
	0.72					
	0.86					
	0.44					

H, *Hydroptila*; O, *Ochrotrichia*; III, 3rd instar; V, 5th instar; pooled, pooled data for 3rd and 5th instars. Proportional similarity (PS) shown with bars under columns.

compared, third-instar *Ochrotrichia* exhibit low similarity (0.44) to all other instars examined. *H. consimilis* third instars are highly similar to fifth instars (0.72–0.86), and the fifth instars of each genus are similar to each other in the proportion use of each trophic resource (0.86).

Discussion

The duration of the *Cladophora* bloom in Bixon Creek and the phenology of the hydroptilid species found there demonstrate that a close insect–algae association exists. Such associations appear to be uncommon among freshwater invertebrate communities (Resh and Houpp 1986), but occur more frequently in marine habitats (e.g., Hay and Fenical 1988, Duffy and Hay 1991, Stachowicz and Hay 1996). Early-instar hydroptilids appeared ≈10 d after *Cladophora* began to proliferate in the stream, and the subsequent decline in the larval community coincided with the crash of the *Cladophora* population. Field data demonstrate that *H. consimilis*, *O. spinosa*, and *O. wojcickiyi* are univoltine in northeastern Ohio. However, laboratory rearings (egg to adult) were conducted with eggs obtained from field-collected adults suggesting the potential existence for bi- or multivoltine populations. *O. wojcickiyi* larvae from some Ohio streams exhibit longer life cycles than those from Bixon Creek (J.B.K., unpublished data). The plasticity of the life cycle of this species allows it to be closely synchronized with the appearance and subsequent senescence of *Cladophora*. Although other algae and Cyanobacteria are present in Bixon Creek, the cells of these organisms are substantially smaller than *Cladophora*. With the exception of diatoms, unicellular food items were consumed infrequently as they rarely occurred in the gut contents analysis, and larger forms were never pierced or otherwise ingested during the many hours of direct observation.

The close association of the microcaddisflies studied and *Cladophora* is further supported by their seemingly specialized morphology. The mandibles of *H. consimilis* are adapted for cell-piercing, whereas those of Bixon Creek *Ochrotrichia* larvae appear to serve a more generalized function. Larvae of the latter genus have mandibles that are well suited for macerating the cell walls of *Cladophora* filaments and for scraping biofilms from algal filaments and riffle rocks. The small size of hydroptilid larvae (Ross 1944, Wiggins 1996) is complimented by their unique mandible morphology. A larger organism would not be able to pierce an algal cell, drain the liquid contents, and leave behind the cellulose-rich cell wall. *Ephemerella needhami* (Ephemeroptera: Ephemerellidae) nymphs are one of the few insect species known to have a strong trophic association with *Cladophora*, but these insects consumed entire cells (McShaffrey and McCafferty 1991). Although energy budgets have not been calculated, it is probably energetically beneficial for hydroptilid larvae to consume the cellular contents only, and avoid digestion of the thick cell walls of *Cladophora*.

The more versatile feeding habits of *Ochrotrichia* may provide an advantage for larvae that inhabit a stream in which the filamentous algae are scoured by a spate or colonize a stream without substantial growths of filamentous algae. In either case, the small larvae could scrape periphyton from riffle rocks to survive. The second situation was observed in a small stream which supported a population of *O. wojcickiyi* (Keiper et al. 1998b). No filamentous green algae were present in the stream, but the cobbles present did have a diatom-rich biofilm and tufts of the red alga *Batrachospermum gelatinosum* (L.) De Candolle (Rhodophyta). Fifth instars were reared to the adult stage by feeding them diatoms, but larvae never consumed the rhodophyte. Furthermore, the versatility of larval feeding by Bixon Creek *Ochrotrichia* is indicated by the high numbers of diatoms found in the guts of fifth instars and the fact that larvae consumed significantly fewer *Cladophora* cells than *H. consimilis* during quantitative laboratory observations.

First-instar *H. consimilis* spent significantly more time crawling/searching than the other larvae tested. Larvae spending most of their time in this behavior expend more energy and feed less. These small larvae probably needed to search for longer periods of time before encountering a cell of appropriate size for piercing. *H. consimilis* larvae exhibited the lowest factor of increase (1.13) between the first and second instars, and larvae did not consume diatoms during laboratory observations. Conversely, *Ochrotrichia* larvae experienced the highest factor of increase during the earliest instars (1.33–1.35). We speculate that *Ochrotrichia* exhibit larger increases in size during early development because of their ability to consume both *Cladophora* cellular contents and lipid-rich diatoms, whereas *H. consimilis* experiences a high factor of increase during later instars because of its specialization on *Cladophora* consumption.

The time to execute one bout of feeding on *Cladophora* exhibited a significant negative relation-

ship with head capsule width. These data illustrate that hypermetamorphosis provides a selective advantage as rapid development to the relatively long duration fifth instar allows hydroptilids to feed on more cells during their larval period. The exceptions to this were the weak trends of *Ochrotrichia* and *Hydroptila* feeding on subapical cells ($R^2 = 0.11$ and 0.45 , respectively). Subapical cells represent the transition zone between small, newly formed cells and the larger mature cells in a filament. The subapical cells are probably variable in size, which is ultimately dependent on age. The unpredictable nature of the size (and thus the volume of protoplast available and the time to pierce the cell wall) of subapical cells may explain the weak relationship between hydroptilid head capsule size and the time to consume these cells and why they were underrepresented in the larval diets.

Close examination of the life histories of *H. consimilis* and *Ochrotrichia* spp. show that they clearly overlap spatially and temporally in Bixon Creek. However, data obtained on morphology of the mandibles, gut contents analysis, and behavior show that trophic resource use was dissimilar. The proportional similarity for the pooled data indicate high similarity in feeding habits between *Hydroptila* and *Ochrotrichia*. It was crucial to look at individual instars to elucidate the mechanisms that reduce the niche overlap of the two genera in Bixon Creek. The greatest difference observed was that third-instar *Ochrotrichia* engaged in diatom grazing significantly more than the other larval stages tested.

Resh and Houpp (1986) asked whether specific trophic associations between aquatic insects and their food sources are rare or overlooked. Their study of the *D. angata*-*Lemanea* interactions and the results given here illustrate a relationship similar to that between a monophagous terrestrial phytophagous insect and its host plant. It appears that the Hydroptilidae exhibit some parallel feeding behaviors found in their terrestrial counterparts; in certain genera (i.e., *Dibusa*, *Hydroptila*, and *Ochrotrichia*, but perhaps others as well), close trophic associations and life cycle synchronization exist between the larvae and a particular genus of alga; larvae feeding on filamentous green algae appear to select for certain areas within a filament to consume; and larval morphology (i.e., body size and mandibular structure) is well-adapted for consuming their host alga (e.g., Nielsen 1948, Lepneva 1964, Flint and Herrmann 1976, Ito and Kawamura 1980, Huryn 1985, Wells 1985, Resh and Houpp 1986, Keiper and Foote 1998, 1999). Further studies of life histories, larval feeding habits, and host alga selection by algivorous Hydroptilidae are needed to strengthen and/or modify these conclusions.

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